



Spatial Analysis Of Groundwater Quality And Its Suitability For Drinking Purpose In Sargodha City

Fatima Iqbal¹, Umer Younis², Omar Riaz³, Muhammad Umar Farooq⁴, Muhammad Haseeb⁵

¹ Department of Earth Sciences, University of Sargodha, Sargodha

Email: iqbalfatimao30@gmail.com

² Department of Earth Sciences, University of Sargodha, Sargodha

Email: umer.younis@uos.edu.pk

³ Department of Earth Sciences, University of Sargodha, Sargodha

Email: omar.riaz@uos.edu.pk

⁴ Assistant Professor, Govt. Graduate College Bhalwal

⁵ Department of Earth Sciences, University of Sargodha, Sargodha

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Corresponding author:

Umer Younis

Email:

umer.younis@uos.edu.pk

ABSTRACT

Groundwater is the principal source of drinking water for Sargodha Tehsil, Punjab, Pakistan, but its quality is increasingly influenced by urbanization, agricultural activities, and changing land-use practices. This study aimed to provide a comprehensive spatial assessment of groundwater quality, evaluate its suitability for drinking, and identify areas vulnerable to potential health risks. The research employed an integrated methodology combining physicochemical analysis, statistical techniques, Geographic Information System (GIS)-based spatial analysis, Groundwater Quality Index (GWQI), Human Health Risk Assessment (HHRA), and household survey information.

A total of 200 groundwater samples were systematically collected across Sargodha Tehsil and analyzed for nine physicochemical parameters, including pH, TDS, EC, turbidity, total hardness, chloride, sulphate, fluoride, and nitrate. The analytical results were compared with World Health Organization (WHO) drinking-water guidelines to assess groundwater suitability. Descriptive statistics, frequency distribution, box-and-whisker plots, and Pearson correlation analysis were applied to determine the variability and relationships among the investigated parameters. The results showed considerable variation in groundwater quality, with localized areas exhibiting elevated concentrations of several parameters, particularly TDS and nitrate.

The georeferenced groundwater-quality data were further analyzed in ArcGIS using Spherical Kriging interpolation to generate spatial distribution maps. The maps revealed pronounced spatial heterogeneity and identified localized groundwater-quality hotspots rather than uniform contamination across the study area. A weighted arithmetic Groundwater Quality Index (GWQI) was subsequently calculated using WHO permissible limits to provide an integrated assessment of drinking-water suitability. The resulting GWQI classification identified areas with comparatively good groundwater quality as well as distinct zones where groundwater quality was poor and unsuitable for drinking.

Human Health Risk Assessment was also conducted to evaluate potential non-carcinogenic health risks associated with groundwater consumption through the ingestion pathway. Hazard Index (HI) values were estimated for both adults and children. The assessment indicated comparatively higher potential health risks for children, particularly in areas where contaminant concentrations were elevated. In addition, household survey data regarding drinking-water sources, bore depths, and water-treatment practices were incorporated to provide a broader understanding of groundwater use and community exposure.

Overall, the study demonstrates significant spatial variability in groundwater quality across Sargodha Tehsil, with specific zones requiring greater attention and management. The integration of hydro chemical analysis, GIS-based Spherical Kriging, GWQI, HHRA, and household assessment provides a robust framework for identifying vulnerable areas and supporting evidence-based groundwater management. The study recommends targeted groundwater-quality monitoring, zone-specific water-treatment measures, improvement of drinking-water infrastructure, public awareness, and sustainable groundwater governance to reduce potential health risks and protect groundwater resources in Sargodha Tehsil.

Introduction

Background of the Study

Groundwater, the water that percolate beneath the Earth's surface and pack the cracks and pore in soil and rock, is one of humanity's most vital, yet often invisible, resources (). It is not just a secondary source; it makes up approximately 99% of all the liquid freshwater on our planet that isn't frozen in glaciers and ice caps. This hidden reserve is the backbone of global water security, sustaining human life, civilization, and the environment in ways that are often taken for granted (Spiro, 2025).

The most direct and essential role of groundwater is providing potable water for billions of people. Globally, about fifty percent of the Earth's population depends on groundwater, either entirely or in part, for their daily needs (Yan, 2025). In many rural areas and regions prone to drought or with limited surface water (like rivers and lakes), groundwater from wells and boreholes is the only accessible source of safe water. This reliance is specifically pronounced in dry climates. The water stored underground in geological formations called aquifers is often naturally filtered, making it less susceptible to pollution and contamination from surface sources than river water (Rizvi, 2025). This relative cleanliness is essential for public health, helping to halt the spread of waterborne sickness and meeting basic human right to clean drinking water. Furthermore, many industrial processes, especially those requiring high-quality water, also depend on this steady and reliable underground supply (Dawoud, 2025).

South Asia is a global leader in groundwater use, withdrawing about 50% of the world's total groundwater. The majority of this water about 89% in some areas like India goes toward irrigated agriculture (Wei, 2025). This massive use, often enabled by private wells and subsidized electricity for pumps (like tube-wells), has been key to the success of the Green Revolution and ensuring food security for billions of people. Without groundwater, food production would be highly vulnerable to the unpredictable nature of the South Asian monsoon.

South Asia is dominated by two actively recharged aquifer systems: the vast Indus Basin (which includes Punjab) and the enormous Ganges-Brahmaputra-Meghna (GBM) Basin (Balasubramanya, 2025). These two huge systems cover much of Pakistan, Northern India, Bangladesh, and Nepal, holding immense reserves of freshwater. These massive alluvial aquifers are constantly being replenished by annual monsoon rains, the extensive network of perennial rivers, and seepage from surface irrigation. Outside of these main basins, there are also numerous smaller, crucial regional aquifer systems in peninsular India and the Himalayan foothills (Rani & Arzjani, 2025). However, the Indus and GBM represent the region's primary, highest-yielding, and most actively recharged groundwater resources. It is the life-support system for over 750 million people.

In Pakistan, the percolation method is a vital strategy for improving groundwater quality, particularly where reliance on boreholes is high and surface water sources (like rivers) are increasingly polluted (Panda, Chidambaram, Ganesh, Biswal, & Alsabti, 2025). Much like in Punjab, the method involves constructing basins or bunds (e.g., check dams) to slow down fast-moving surface water during monsoon rains. By holding the water in place for an extended period, the method allows for maximum seepage into the ground. This process is crucial because it helps address the widespread issue of bacterial and sewage contamination, especially in peri-urban areas where wastewater management is often inadequate (Basharat, 2025).

The groundwater resource in Punjab, often celebrated as India's agricultural heartland and food basket, is currently facing an unprecedented environmental crisis that forms the central focus of this research (Kanwar, Khan, & Shoaib, 2025). The problem is rooted in the success of the Green Revolution era, where the state rapidly shifted its agricultural strategy. Initially, this involved the expansion of canal networks, but this quickly gave way to a heavy reliance on groundwater extraction via tube wells. Today, agriculture consumes a disproportionate share of the total water budget approximately 97% making the sustainability of the aquifer directly synonymous with the sustainability of the state's economy (Rizvi, 2025).

Result of this intensive extraction is a rapid and alarming rate of groundwater depletion. Numerous studies confirm that the annual withdrawal of water from the aquifers significantly surpasses the natural rate of recharge from precipitation and canals (Olusanya, 2025). This unsustainable water mining has led to an official designation of nearly 79% of the state's administrative blocks as "Over-Exploited" by government bodies. The decline is not uniform; the central districts of Punjab, where rice cultivation is most concentrated, are the worst affected, with the water table falling at an average rate of close to half a meter every year. This continuous drop forces farmers to bore deeper and deeper wells, which exponentially increases the energy required for pumping and places an enormous financial burden on the agricultural community, particularly small and marginal landholders (Gebreslassie, 2025).

Beyond the sheer quantity of water, the crisis is compounded by significant issues of water quality and contamination. As the water table descends, farmers are forced to draw water from deeper strata, which may contain naturally elevated concentrations of harmful elements like arsenic and fluoride (Bjornlund, 2025). Simultaneously, the shallow aquifers, which are essential for supplementary recharge, are susceptible to contamination from surface-level activities.

Groundwater challenge in Punjab transcends mere hydrological metrics to become a complex socio-economic and policy challenge. The crisis is structurally reinforced by flawed incentive systems, most notably the provision of subsidized or free electricity for agricultural use and the guaranteed procurement of water-intensive crops at a Minimum Support Price (MSP) (Kamruzzaman et al., 2025). These policies have created a cycle where farmers are financially incentivized to pump groundwater without considering its economic or ecological cost, delaying the urgent adoption of water-saving technologies. The widening gap in access to water where large landholders can afford the capital to drill deep submersible wells while poorer farmers face well failure raises critical questions of equity and social justice (Kishore, Birthal, & Srivastava, 2025).

The quality issue is exacerbated by the problem of groundwater depletion. Punjab, a heavily agrarian region, has an extremely high rate of groundwater extraction, primarily for irrigation (Raina, 2025). This rate often far exceeds the natural rate of recharge from rainfall, leading to an 'over-exploited' status in most blocks. As the water table drops, deeper and potentially poorer quality water is accessed. This depletion also concentrates existing contaminants and, in some cases, mobilizes geogenic contaminants like arsenic and fluoride by changing the water chemistry. This significant drop in water level, observed across most of the state, creates an urgent challenge for long-term sustainability.

A percolation tank or pond is essentially a reservoir built across a natural stream or water channel. When it rains, this structure captures surface runoff that would otherwise be lost. Instead of flowing away, the water slowly infiltrates through the soil and rock layers the process of percolation to recharge the underlying aquifer. This recharge action directly raises the water table, making water more accessible. Critically, as the water moves through the soil layers, it undergoes natural filtration, which helps to reduce the concentration of certain contaminants like nitrates and pesticides before they reach the main groundwater supply (Rizvi, 2025).

This method also indirectly combats salinity and fluoride issues by diluting the existing, highly concentrated groundwater with fresh, filtered rainwater. The large-scale implementation of percolation tanks can create decentralized points for groundwater replenishment, reducing the stress on deep tube wells and improving the overall balance of the local water cycle (Acharya, 2025).

Groundwater crisis in Punjab is characterized by three main types of contamination: Arsenic, Salinity, and Industrial/Bacterial pollution. Central Punjab (Lahore, Faisalabad, Sheikhupura, Gujranwala). These cities face heavy industrial pollution. In Lahore, the water table is falling by nearly 2.5 to 3 feet annually, leading to the intrusion of low-quality water. In Faisalabad and Sheikhupura, industrial waste (effluents) has introduced heavy metals like chromium and lead into the deep aquifers.

Southern Punjab (Multan, Bahawalpur, Rahim Yar Khan, Muzaffargarh), this region is a "hotspot" for Arsenic contamination, which occurs naturally but is worsened by over-pumping. In Muzaffargarh, studies show high levels of arsenic poisoning in children. Furthermore, 60-70% of groundwater in the south is naturally saline (brackish), making it unfit for both drinking and agriculture.

Specific Clusters (Kasur & Sargodha), Kasur is notorious for high fluoride and chromium levels due to its leather tanneries. In Sargodha, recent reports highlight high Total Dissolved Solids (TDS) and nitrates, largely due to agricultural runoff and poor waste management.

- **Punjab Saf Pani Company (PSPC)**, this government-led project has installed hundreds of water filtration plants (Ultra-Filtration and Reverse Osmosis) in tehsils like Chak Jumrah and various districts in Southern Punjab.
- **Abe-Pak Project** managed by the Punjab Aab-e-Pak Authority, this project aims to install filtration plants across the province to ensure the provision of clean water at the community level.
- **Asian Development Bank (ADB) Projects**, ADB is investing nearly \$250 million specifically in Sialkot and Sahiwal to improve municipal infrastructure and prevent sewage from mixing with drinking water lines.

The Sargodha region's groundwater is stored within an unconfined alluvial aquifer a large underground layer of sand, silt, and clay deposited by the Indus tributaries over millennia (Dawoud, 2025). The aquifer is generally thick, offering significant storage

potential. Its lower boundary is often defined by a buried, relatively impermeable bedrock structure, such as the Delhi-Shahpur Ridge, which influences groundwater flow.

Study Area

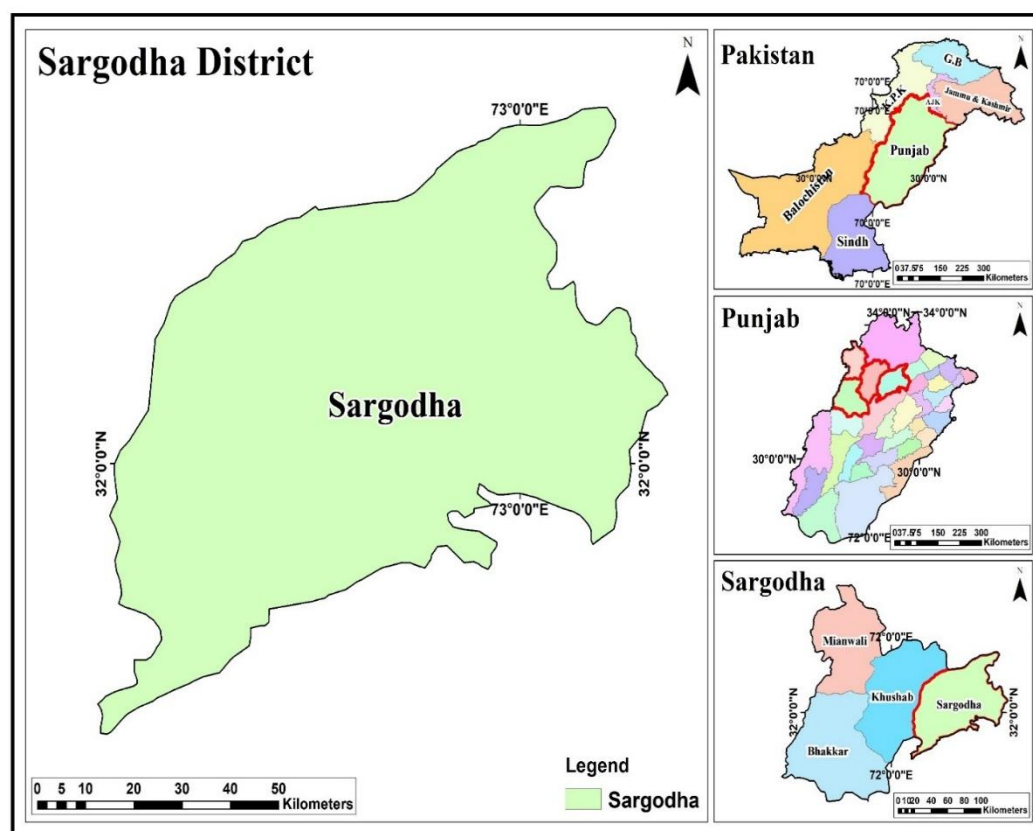
The study area for this research is Sargodha City, located in the Sargodha District of Punjab, Pakistan (figure 1). The city is situated in the Chaj Doab, the fertile plain between the Jhelum and Chenab rivers (Masood, 2024). Sargodha is a significant urban and administrative center, with a population of over 975,000 (2023), and it serves as the hub for a largely agricultural region known for its high-quality citrus production.

The city's geography is characterized by its largely flat, alluvial plains at an average elevation of 190 meters above sea level (MALIK, 1958). This topography makes the groundwater aquifer a primary source of water for both domestic and agricultural use. The area experiences a semi-arid climate with hot summers and moderate winters. The rapid urbanization and industrial growth of Sargodha have placed significant pressure on its groundwater resources, making them susceptible to contamination

from both natural and anthropogenic activities.

Boundaries of Sargodha City's municipal corporation will serve as the geographical scope. The research will involve collecting groundwater samples from various locations within these boundaries, including residential, commercial, and industrial zones, to capture the spatial variation of water quality. This approach will allow for a comprehensive GIS-based analysis to identify areas where groundwater is most vulnerable to pollution (Arooj, 2025).

Figure 1 Geographical location of study area map, Sargodha city



Methodology

The initial objective of the research focused on establishing the current hydro geochemical characteristics of groundwater within the Sargodha City study area. This involved a systematic field sampling campaign to collect representative groundwater samples

from established bore wells and pumping stations across the city. The sampling locations were determined using a GIS-based approach to ensure adequate spatial coverage, incorporating factors such as land-use type and well density. After collection, each sample was immediately transported to a certified laboratory under controlled conditions for comprehensive analysis. The analysis focused on key physicochemical parameters, including pH, Electrical Conductivity EC, Total Dissolved Solids TDS, turbidity and major ions such as, nitrate, chloride, fluoride and sulfate. These results were then compiled into a central database and subjected to initial statistical analysis to identify fundamental trends, dominant water types, and the initial chemical composition of the sargodha city aquifer system.

The research begins with a comprehensive survey of the Sargodha district. The area's diverse land use ranging from dense urban centers to expansive citrus orchards a stratified random sampling approach will be used. This ensures that samples are collected from various environments to capture different chemical signatures. Using a grid-based system, approximately sampling sites will be identified, focusing on functional tube wells and hand pumps. Each location's coordinates will be precisely recorded using a GPS device to allow for accurate mapping of chemical variations across the district.

Before collecting water, each source will be pumped for several minutes to remove stagnant water from the pipes, ensuring the sample reflects the actual condition of the aquifer. High-density polyethylene (HDPE) bottles will be used for storage, these will be pre-cleaned and rinsed multiple times with the target groundwater at the site.

Once transported to the laboratory, the samples will undergo rigorous chemical testing. The analysis focuses on major anions (including Chloride, Sulfate, and Nitrate). These will be measured using standard methods like Flame Photometry and Titration. To ensure the data is accurate, a Charge Balance Error (CBE) calculation will be performed for every sample. If the difference between the positive and negative ions exceeds 5%, the sample will be re-analyzed to ensure the highest level of data integrity.

The final phase involves interpreting the laboratory results to understand the "personality" of Sargodha's groundwater. The data will be integrated into GIS software to create visual maps, allowing for a clear assessment of which areas in Sargodha meet safety standards and which are facing chemical degradation. Distribution analysis (frequency histograms), correlation analysis (heat map), correlation scatter plots, Box-and-Whisker Analysis, variability comparison across parameters, sample wise trend analysis and human health risk assessment (HHRA) were conducted that helps to identify the water's origin.

The second objective rely on the utilization of Geographic Information System GIS techniques to investigate the spatial distribution of key water quality parameters and their relationship with Land Use/Land Cover (LULC) patterns across Sargodha City. The analysis was heavily reliant on the georeferenced groundwater sample data collected in first objective, with each laboratory result precisely linked to its GPS location. Using GIS software, continuous spatial distribution maps for critical parameters (such as TDS and nitrates) were generated through Kriging interpolation, a geostatistical modeling technique well-suited for environmental data. This interpolation creates a predictive surface that estimates parameter values even between sample points. A LULC map of the study area was created or acquired, classifying areas into categories like residential, commercial, agricultural, and industrial. The primary technique involved an overlay analysis within the GIS environment, which allowed for the visual and statistical comparison of the interpolated water quality maps against the LULC zones. This comparative analysis helped quantify and determine if specific land use patterns were significantly associated with areas of high or low concentration of water quality contaminants. This objective relied on integrating the sample data with the existing spatial LULC model.

The analysis depends significantly on this georeferenced dataset, ensuring that laboratory findings for parameters like Total Dissolved Solids (TDS) and nitrates are positioned exactly according to their collection points. In the GIS environment, continuous spatial distribution maps are produced for these critical parameters through the application of Kriging interpolation. This geostatistical approach is particularly appropriate for environmental datasets, as it accounts for spatial autocorrelation and variability inherent in groundwater quality measurements.

Kriging interpolation functions by generating a predictive surface that extends beyond the discrete sample locations, providing estimated values for water quality parameters in unsampled areas. This results in smooth, continuous maps that reveal broader spatial trends across Sargodha City, highlighting zones of elevated or reduced contaminant levels.

Both visual inspection and statistical evaluation become possible, allowing identification of patterns where high TDS or nitrate levels coincide with particular LULC categories. The visual component reveals clustered hotspots or gradients, while statistical measures assess the strength of associations between land use patterns and contaminant distributions.

This comparative examination quantifies the extent to which certain land uses such as intensive agricultural practices or industrial activities correspond to areas of higher water quality contamination, or conversely, where residential or less disturbed zones exhibit lower concentrations.

The third objective of the research was to evaluate the suitability of the groundwater in Sargodha City for human consumption. This objective was highly dependent on the laboratory analysis of the groundwater samples, as the concentration of every key parameter determines compliance with health standards. For this evaluation, a Ground Water Quality Index (GWQI) was calculated. The GWQI served as the primary model for this objective, converting the complex array of chemical results e.g., TDS, Nitrates, Chloride into a single, standardized numerical score. This calculation involved assigning a relative weight to each parameter based on its perceived impact on health and then normalizing the concentration against established drinking water standards e.g., WHO or local Pakistani standards. Following the calculation of the GWQI for each sampling point, GIS techniques were reapplied. Using the spatial analysis tools, a continuous GWQI suitability map was interpolated, visually classifying areas of the city into different quality categories e.g., excellent, good, poor, unsuitable.

It is essential to recognize how central the laboratory analysis phase was. The evaluation of groundwater suitability for human consumption hinged entirely on accurate measurements from these samples. Each key parameter's concentration played a critical role in assessing whether the water met health standards. The introduction of the Ground Water Quality Index (GWQI) provided a robust framework, acting as the primary model to simplify the interpretation of multiple chemical parameters, such as TDS, Nitrates, and Chloride, by transforming them into one comprehensive score. After computing GWQI per point, the reapplication of GIS tools facilitated the creation of an interpolated map, providing a spatial overview and clear categorization of quality zones across the city.

The last objective focused on creating a long-term, practical tool for groundwater quality management in Sargodha City using a GIS-based approach. It directly incorporated results from the previous objectives, particularly drawing from the GWQI suitability map developed in third objective and the linkages established between land use patterns and contamination sources. No additional fieldwork was required no new samples were collected, and no primary questionnaires were administered. The framework was constructed entirely from the existing spatial database already built in the GIS environment, which served as the foundation for defining all essential components.

Key elements of the framework included proposing an optimal sampling network prioritized according to high-risk zones highlighted by the GWQI map. It also emphasized the routine incorporation of fresh laboratory results into the GIS database, allowing for the generation of updated comparative maps. These maps would immediately highlight regions exceeding drinking water standards, thus facilitating precise management actions and optimized resource distribution to safeguard public health.

Groundwater Water Quality Index (GWQI) Formulation

To integrate the individual parameters into a single comprehensive indicator of water quality, the Weighted Arithmetic Water Quality Index (WAWQI) method (Brown et al., 1972; Horton, 1965) was used. This method assigns a relative weight to each parameter based on its importance/WHO standard, and combines the parameters into a single index value (GWQI).

Steps and Formulas for GWQI Calculation

Step 1: Unit Weight (W_i)

Each parameter is assigned a unit weight inversely proportional to its WHO recommended standard:

$$W_i = K / S_i$$

where S_i = WHO standard (permissible limit) value of the i^{th} parameter, and K = proportionality constant.

Step 2: Proportionality Constant (K)

$$K = 1 / \sum (1 / S_i)$$

Step 3: Quality Rating Scale (Q_i)

The quality rating for each parameter is calculated as:

$$Q_i = [(V_i - V_o) / (S_i - V_o)] \times 100$$

where V_i = mean observed (measured) value of the i^{th} parameter, S_i = WHO standard value of the i^{th} parameter, and V_o = ideal value of the parameter in pure water ($V_o = 0$ for all parameters, except pH, where $V_o = 7$).

Note: Many published groundwater theses simplify this to $Q_i = (V_i / S_i) \times 100$ since $V_o = 0$ for most parameters. For pH, the ideal-value form above must be used.

Step 4: Overall Groundwater Water Quality Index (GWQI)

$$GWQI = \Sigma (W_i \times Q_i) / \Sigma W_i$$

Table 1: GWQI Classification Scale (established by Brown et al. and ScienceDirect.com Studies)

GWQI Value	Water Quality Category
0 – 25	Excellent
26 – 50	Good
51 – 75	Poor
76 – 100	Very poor
> 100	Unsuitable for drinking

Worked Example (Nitrate Parameter, for Illustration)

- S_i (WHO standard) = 50 mg/L
- V_i (mean observed) = 6.08 mg/L
- $V_o = 0$

$$Q_i = (6.08 - 0) / (50 - 0) \times 100 = 12.16$$

$$W_i = K / 50 \text{ (K is calculated using all nine parameters' } S_i \text{ values)}$$

This Q_i and W_i value is then combined with the corresponding values of the remaining eight parameters using the GWQI formula in Step 4 to obtain the final composite index for each sample/site.

Figure 2 :GWQI Trends Across 200 Groundwater Samples

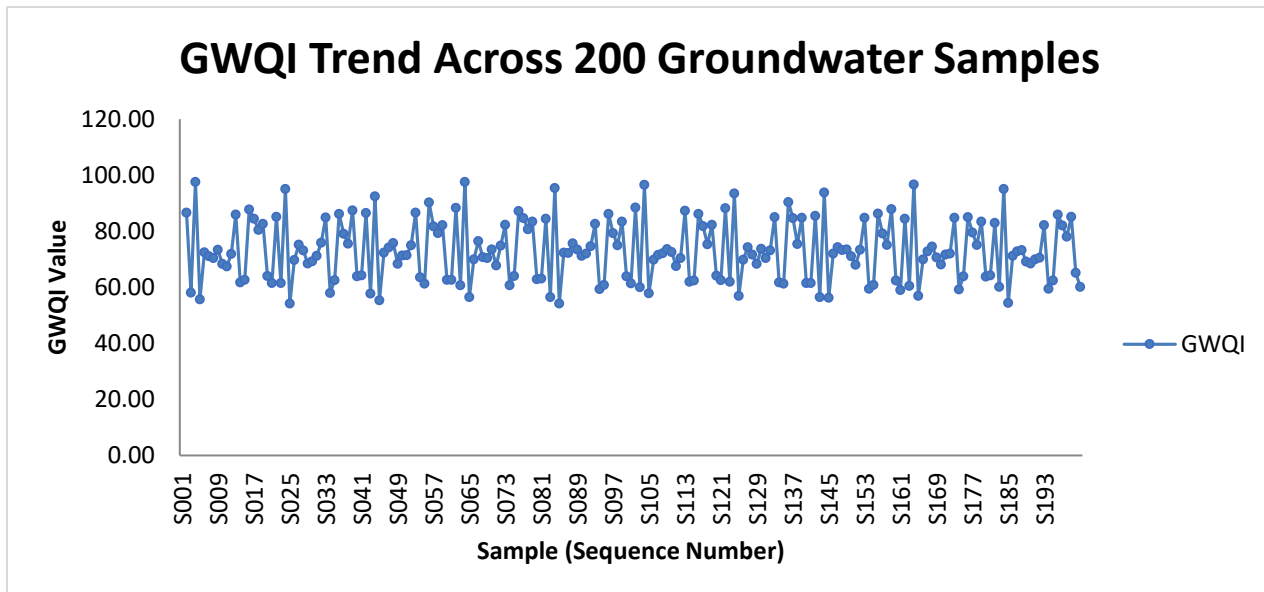


Figure 3: Water Quality Classification, Sample Count

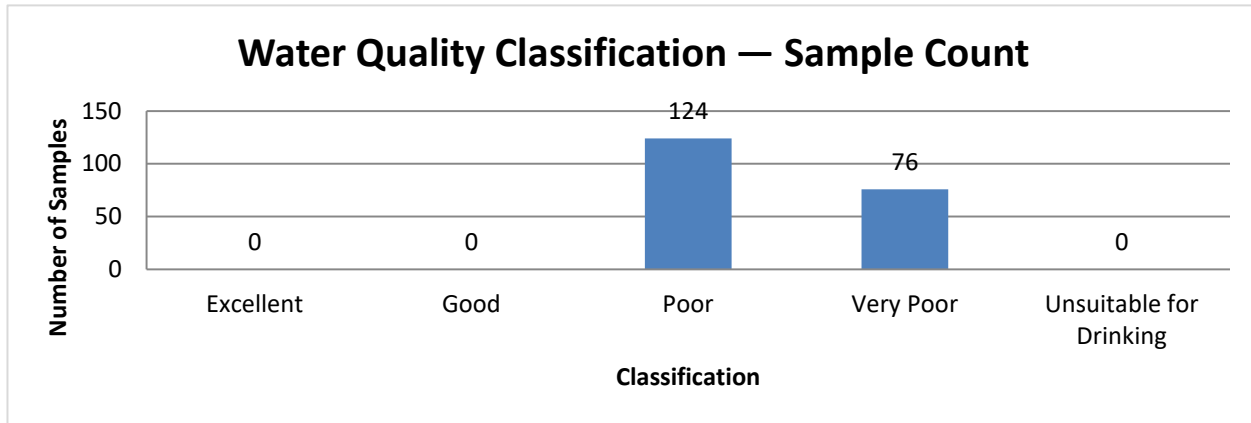


Figure 4: Water Quality Classification, Percentage Share

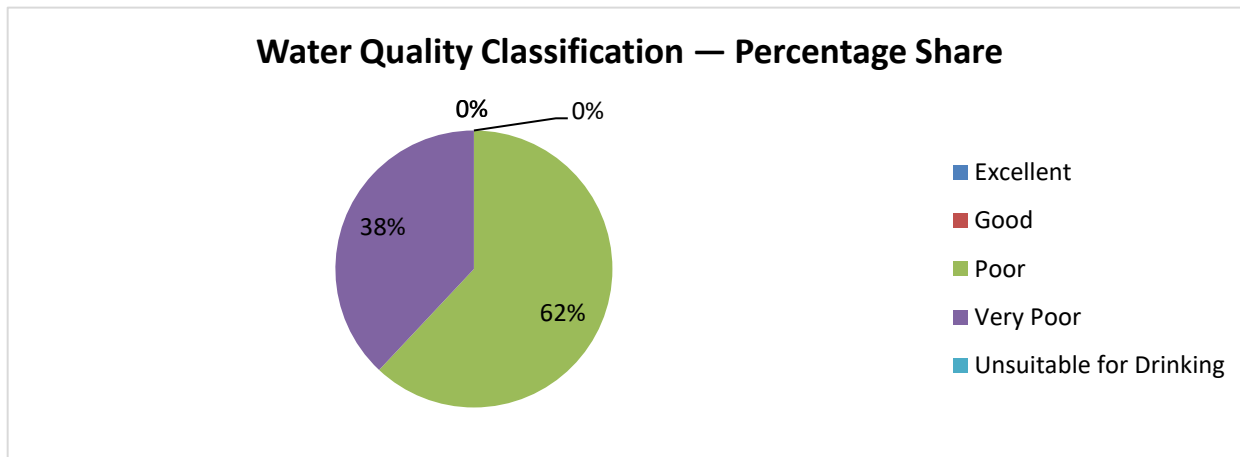


Figure 5: Shows Mean Quality Rating (qi) by Parameter, Pollution Contribution

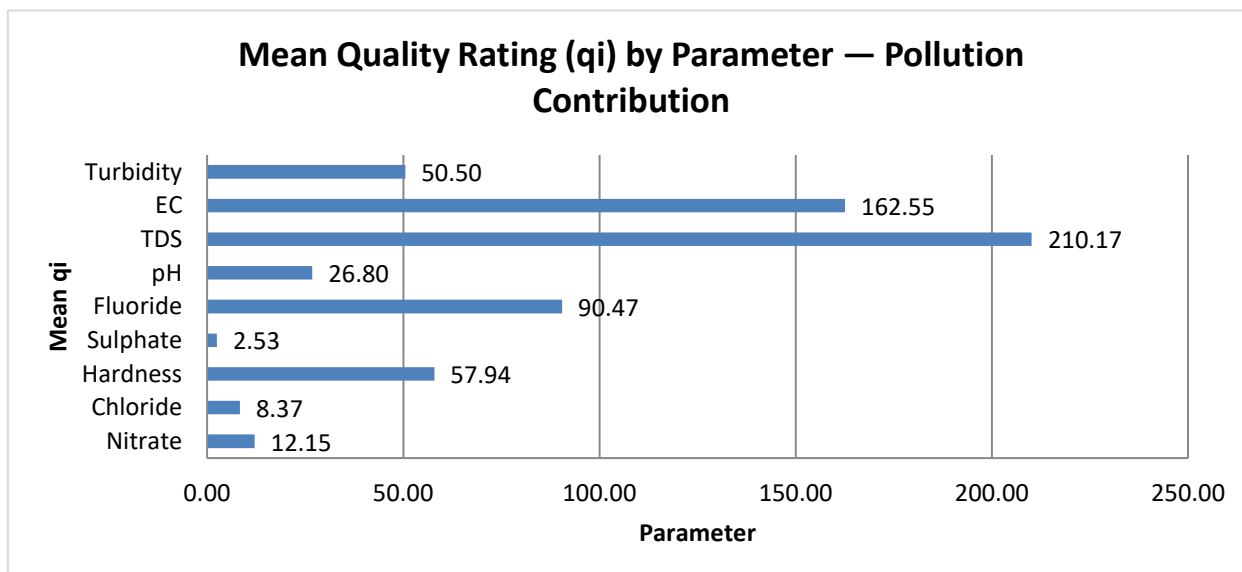


Figure 5 shows the mean quality rating (qi) of each groundwater quality parameter. TDS (210.17) exhibited the highest quality rating, followed by EC (162.55) and fluoride (90.47), indicating that these parameters contributed the most to groundwater quality deterioration. In contrast, sulphate (2.53), chloride (8.37), and nitrate (12.15) showed the lowest quality ratings, reflecting their relatively minor contribution to overall groundwater pollution. Overall, TDS and EC were the dominant parameters affecting groundwater quality in the study area.

Table 2: Shows Groundwater Quality Classification Distribution

Water Quality Classification Distribution

Classification	GWQI Range	Sample Count	Percentage
Excellent	0 – 25	0	0.0%
Good	26 – 50	0	0.0%
Poor	51 – 75	124	62.0%
Very Poor	76 – 100	76	38.0%
Unsuitable for Drinking	> 100	0	0.0%
Total		200	100.0%

Table 3: WHO Drinking-Water Standards & Computed Unit Weights

Parameter	Unit	WHO Standard (Sn)	Ideal Value (Vi)	1 / Sn	Unit Weight (wi = K / Sn)
Nitrate (NO ₃ ⁻)	mg/L	50.00	0.00	0.020000	0.019685
Chloride	mg/L	250.00	0.00	0.004000	0.003937
Total Hardness (as CaCO ₃)	mg/L	500.00	0.00	0.002000	0.001969
Sulphate	mg/L	250.00	0.00	0.004000	0.003937
Fluoride	mg/L	1.50	1.00	0.666667	0.656181
pH	-	8.50	7.00	0.117647	0.115797
TDS	mg/L	1000.00	0.00	0.001000	0.000984
EC	uS/cm	1500.00	0.00	0.000667	0.000656
Turbidity	NTU	5.00	0.00	0.200000	0.196854
Σ (1/Sn)				1.015980	
K = 1 / Σ(1/Sn) (proportionality constant)				0.984271	
Check: Σ (wi) should equal 1.000000					1.000000

Table 3 presents the WHO drinking-water quality standards, the corresponding ideal values, and the computed unit weights (wi) used in the calculation of the Water Quality Index (WQI). The WQI method assigns a relative importance (weight) to each water quality parameter based on its permissible standard value.

The parameters included are Nitrate (NO₃⁻), Chloride, Total Hardness (as CaCO₃), Sulphate, Fluoride, pH, Total Dissolved Solids (TDS), Electrical Conductivity (EC), and Turbidity. For each parameter, the WHO standard value (Sn) and ideal value (Vi) are specified. The reciprocal of the standard value (1/Sn) is calculated to determine the relative significance of each parameter.

A proportionality constant (K) is then computed using:

$$K = \frac{1}{\sum(1/S_n)}$$

Based on the calculated value of $K = 0.984271$, the unit weight for each parameter is determined using:

$$w_i = \frac{K}{S_n}$$

The calculated unit weights indicate the contribution of each parameter to the overall WQI. Parameters with lower permissible limits receive higher weights because they have a greater impact on drinking-water quality. For example, **Fluoride** ($w_i = 0.656181$) and **Turbidity** ($w_i = 0.196854$) have the highest weights, reflecting their greater significance in assessing water quality, whereas **TDS** ($w_i = 0.000984$) and **EC** ($w_i = 0.000656$) have relatively lower weights due to their higher permissible limits.

The sum of all unit weights equals **1.000000**, confirming that the weighting system has been correctly normalized. These normalized weights are subsequently used in WQI calculations to evaluate the overall suitability of water for drinking purposes according to WHO guidelines.

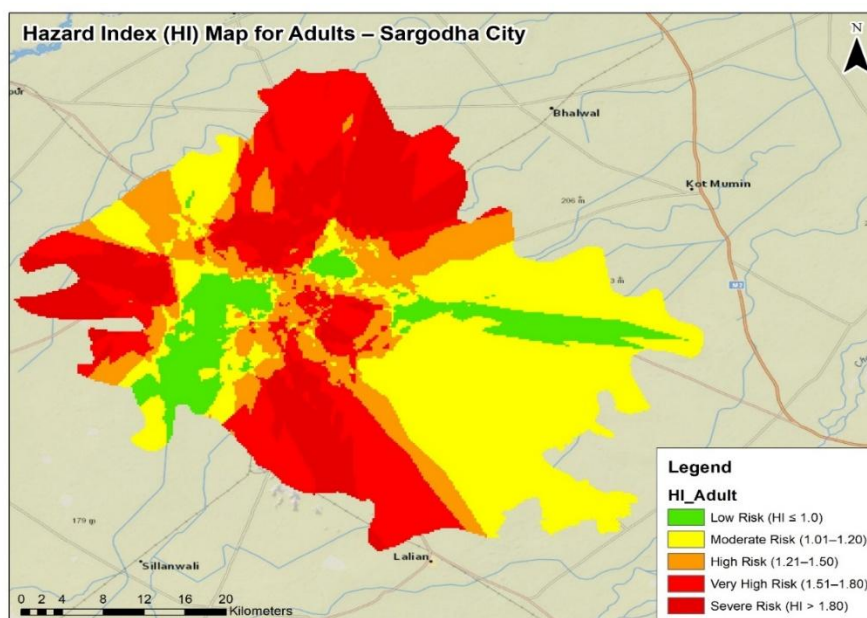
Human Health Risk Assessment (HHRA)

The figures presented in human health risk assessment (HHRA) depict a household water-use environment characterized by heavy reliance on shallow, privately operated bores, a widespread absence of household-level water treatment, a considerable burden of self-reported gastrointestinal, dermatological, and skeletal health complaints and a marked disparity in both health outcomes and environmental exposure between hotspot and lower-risk localities.

Spatial Distribution of the Hazard Index

To translate the household-level water-quality and health findings into a spatial format suitable for planning purposes, a non-carcinogenic Hazard Index (HI) was computed for each sampling grid across Sargodha city and interpolated across the study area separately for adult and child receptors. The Hazard Index expresses the ratio of estimated daily contaminant intake, derived from measured physicochemical parameters at each grid point, to the corresponding reference dose for that contaminant, summed across all parameters considered. A value at or below unity indicates that cumulative exposure remains within the tolerable threshold for that population group, whereas values exceeding unity indicate a hazard burden that may pose an appreciable risk of adverse non-carcinogenic health effects. The resulting surfaces were classified into five risk categories, ranging from low risk ($HI \leq 1.0$) to severe risk ($HI > 1.80$), and are presented as thematic maps in Figures 6 and 7.

Figure 6: Spatial Distribution of the Hazard Index (HI) for Adults, Sargodha City



The adult Hazard Index map (figure 6) reveals a heterogeneous risk surface across Sargodha city. Zones classified as very high or severe risk, shown in red, dominate the northern periphery of the urban core as well as a broad arc extending through the southern and southwestern parts of the study area, coinciding with localities identified elsewhere in this research as contamination hotspots. A cluster of low-risk grids, shown in green, is concentrated in the central and west-central part of the city, while a distinct band of moderate-risk territory extends eastward toward the outskirts of the built-up area. High-risk zones, shown in orange, form a transitional fringe between the low-risk core and the surrounding very-high and severe-risk belts. The overall pattern suggests that adult exposure risk is not uniformly distributed across the city but instead follows a roughly concentric gradient, with comparatively safer conditions prevailing near the city center and risk intensifying toward the northern and southern margins, where reliance on shallow, untreated groundwater is most pronounced.

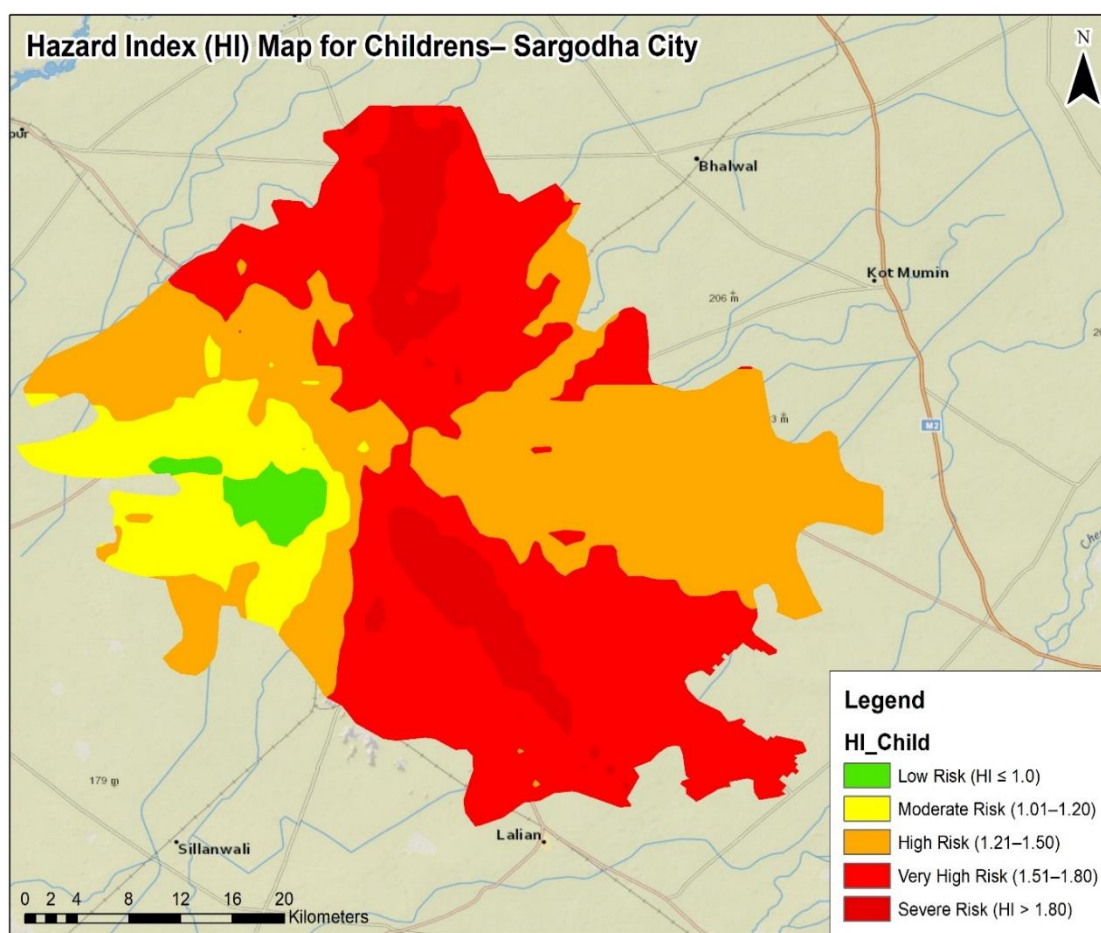


Figure 7: Spatial Distribution of the Hazard Index (HI) for Children, Sargodha City

The corresponding map for children (figure 7) displays a substantially more severe and spatially extensive hazard pattern than that observed for adults. Very high and severe-risk zones, shown in red and dark red, cover the majority of the eastern, northern, and southern portions of the study area, while high-risk zones, shown in orange, occupy much of the remaining built-up extent. Only a small, localized pocket of low-risk territory persists in the western part of the city, surrounded by a narrow ring of moderate-risk grids. This pronounced expansion of high and severe-risk territory relative to the adult map is consistent with the well-established toxicological principle that children, owing to their lower body mass, higher water intake relative to body weight, and ongoing physiological development, experience a proportionally greater dose of ingested contaminants than adults exposed to the same water source. Considered jointly, Figures 6 and 7 indicate that children across nearly the entire study area, and adults across a substantial majority of it, face a cumulative non-carcinogenic hazard burden that exceeds the tolerable threshold,

underscoring the urgency of targeted water-quality remediation, particularly in the northern and southern zones where both maps concur in indicating severe risk.

Conclusions and Recommendations

Conclusion and recommendations comprises of summary of key findings of all parameters, synthesizing the spatial-hydro geochemical-health nexus and policy recommendations for urban water management in Sargodha.

Summary of Key Findings

This study evaluated the spatial distribution and chemical quality of groundwater in the study area to assess its suitability for consumption. By analyzing key chemical parameters and employing Kriging (Spherical) interpolation for spatial mapping, several critical insights regarding local water quality were uncovered. The concentrations of Nitrate, Chloride, and Sulphate were all found to be well below the permissible thresholds established by the World Health Organization (WHO). These low values indicate that the study area currently faces negligible risk from agricultural runoff, sewage contamination, or salinity intrusion.

In contrast, Fluoride levels averaged 1.45 mg/L, with maximum concentrations reaching 1.62 mg/L. Because these values approach and occasionally exceed the WHO limit of 1.50 mg/L, there is a localized threat of dental and skeletal fluorosis for the local population, highlighting the need for targeted spatial monitoring.

The geospatial interpolation maps successfully delineated the variations of physical water properties across the region, pH ranged from 7.22 to 7.73, indicating stable, slightly alkaline conditions, turbidity remained low, fluctuating between 2.08 NTU and 3.43 NTU, total hardness spanned from 197.3 to 366.8 mg/L, showing moderate to high mineral content.

TDS and EC both parameters showed elevated levels, with Total Dissolved Solids ranging from 1656.68 ppm to 2538.47 ppm and Electrical Conductivity varying between 1962.77 uS/cm and 2852.3 uS/cm, pointing to high overall mineralization in the groundwater.

The groundwater is generally safe from organic pollution and common industrial/agricultural anions, its high mineralization (TDS/EC) and elevated fluoride levels pose potential long-term health and usability challenges. The Kriging interpolation maps generated in this research serve as a valuable tool for resource management, identifying specific zones where water treatment particularly de fluoridation is necessary to ensure safe drinking water for the community.

Main Conclusions (Synthesizing the spatial-hydro geochemical-health nexus)

The ultimate objective of this research was to move beyond isolated data points and understand how the physical movement of water, chemical contaminants, and human lives intersect across the landscape of Sargodha City. By weaving together laboratory results, advanced geographic modeling, and household surveys, this study confirms that groundwater quality is deeply intertwined with both the local environment and public health.

The Interconnected Nexus

The core conclusion of this thesis is that the groundwater in Sargodha City cannot be viewed as a natural resource, but must be understood as a dynamic system where geology and human activity directly impact human biology. The laboratory analysis of the nine core parameters specifically variables like Total Dissolved Solids (TDS), Electrical Conductivity (EC), nitrates, and fluorides revealed that a significant portion of the urban aquifer exceeds safe consumption thresholds. The true meaning of these chemical numbers only becomes clear when viewed through a spatial lens. The Kriging interpolation models proved that these contaminants are not scattered randomly, they form distinct geographic hazard zones, heavily influenced by local soil characteristics, dropping water tables across the 200 sampling points, and poor urban drainage systems.

Connecting the Environment to Human Health

The final link in this spatial-health nexus is the direct relationship between these contaminated geographic zones and the vulnerability of the local population. By overlaying the spatial Hazard Index maps with the data collected from the 20-variable public survey, this establishes a clear, undeniable pattern: households living within high-risk geographic grids suffer from a disproportionately higher burden of health issues. Chronic complaints such as gastrointestinal distress, bone and joint pain, and early signs of dental issues perfectly align with the spatial hotspots of high sulfate and fluoride concentrations. The synthesis proves that this health nexus is deeply unfair; children are at a much higher physiological risk than adults in the exact same

neighborhoods due to their lower body weight and higher relative water intake. The research concludes that without immediate, spatially targeted interventions, the degrading hydro geochemical profile of Sargodha will continue to drive a quiet but serious public health crisis across the city's most vulnerable communities.

Policy Recommendations for Urban Water Management in Sargodha

The empirical findings of this study provide a critical roadmap for local regulatory bodies, including the Sargodha Development Authority (SDA), the Water and Sanitation Agency (WASA), and public health departments. To mitigate the growing challenges of groundwater degradation and protect the local population, the following structural and strategic interventions are strongly recommended.

Implementation of Zone-Based Pumping and Regulatory Frameworks

The spatial maps generated via Kriging interpolation clearly demonstrate that groundwater degradation is not uniform across Sargodha City rather, it is concentrated in distinct geographic hotspots. Local authorities must enforce strict zoning laws regarding water extraction. In sectors where the water table is drastically dropping or where chemical contaminants like nitrate and fluoride exceed safe limits, the installation of new, shallow domestic tube wells should be legally restricted. The municipal government should regulate deep-aquifer extraction and monitor industrial or agricultural water usage in the periphery of the city to prevent further depletion and lateral movement of contaminated plumes.

Deployment of Targeted, Grid-Specific Water Treatment Infrastructure

Different areas of Sargodha exhibit varying hydro geochemical characteristics, water treatment strategy is highly inefficient. Policy planners should utilize the GIS risk maps produced in this research to deploy targeted water purification infrastructure. Community grids showing critically high levels of Total Dissolved Solids (TDS) and Fluoride, the government should prioritize the installation of localized, low-cost Reverse Osmosis (RO) plants or specialized activated alumina filters. In areas marked by high turbidity and microbiological risks, community-level chlorination and rapid sand filtration units should be established to ensure the immediate supply of pathogen-free drinking water.

Upgradation of Urban Infrastructure and Source Protection

The presence of elevated nitrate levels in specific urban clusters strongly points toward the leaching of domestic sewerage and poor solid waste management into the shallow unconfined aquifer. There is an urgent need to rehabilitate Sargodha's aging wastewater infrastructure. Municipal bodies must ensure the complete physical separation of open sewage channels from municipal drinking water pipelines to eliminate cross-contamination. Strict environmental regulations must be enforced on local small-scale industries and workshops, mandating the proper treatment of their chemical effluents before disposal, thereby cutting off the source of hazardous groundwater recharge.

Integration of Public Surveys into Community Health and Awareness Programs

The socio-environmental data collected through the 20-variable public perception survey indicates a significant gap between water quality awareness and actual household health burdens. The Health Department, in collaboration with local educational institutions, should launch targeted public awareness campaigns in identified high-risk zones. Residents must be educated on simple, cost-effective household water treatment methods, such as boiling and basic filtration. A transparent, public-facing groundwater quality monitoring dashboard should be established by the city management, allowing citizens to check the safety status of the water in their respective union councils, fostering a community-driven approach to water safety.

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